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CS 230 : Discrete Computational Structures

Summer Semester, 2025

ASSIGNMENT #1

Due Date: Friday, May 30

Suggested Reading: Rosen Sections 1.1 - 1.3; LLM Sections 1.1, 3.1 - 3.4, Lecture Notes Logic 4 - 7; Rosen Sections 1.4 - 1.6; LLM Chapter 3

These are the problems that you need to hand in for grading. Always explain your answers and show your reasoning.

1. **[6 Pts]** Consider the statement “If you are the bad guy, then you break the rules.” Provide the converse, inverse and contrapositive of the statement. *Your statements should be in English and not in logical form.*

Converse: If you break the rules, then you are the bad guy.

Inverse: If you aren't the bad guy, then you don't break the rules.

Contrapositive: If you don't break the rules, then you aren't the bad guy.

2. [6 Pts] Translate the following English sentences into logic. First, define your basic propositions and use logical operations to connect them.

(a) If n is prime, then $n = 2$ or n is odd.

P: n is prime

T: $n = 2$

O: n is odd

$$P \rightarrow (T \vee O)$$

(b) The differentiability of f is sufficient for f to be continuous and smooth.

D: f is differentiable

C: f is continuous

S: f is smooth

$$D \rightarrow (C \wedge S)$$

(c) Being a good programmer and having good communication skills are necessary to get a good job.

J: Getting a good job

P: Being a good programmer

C: Having good communications skills

$$J \rightarrow (P \wedge C)$$

3. [10 Pts] Use logical reasoning or truth tables to solve the following puzzle:

Five friends disagree on whether to play video games or basketball. Either Arya or Bran, or both, want to play video games. Catelyn and Davos disagree on what they want to play. If Eddard plays video games, then so does Catelyn. Arya and Davos will play the same game. If Bran plays video games, then so do Arya and Eddard. Who is playing what?

Define five basic propositions a, b, c, d, e and use these to describe the five statements above. Then use logical reasoning to derive the truth of each of those basic propositions.

You should use logical reasoning here, truth table will not be accepted.

Hint: Let a be 'Arya play video games'. This means that $\neg a$ is 'Arya plays basketball'. Start by assuming that Bran plays video games and get a contradiction. That tells you that Bran plays basketball, i.e. $\neg b$. Now, deduce whether a, c, d and e are true or false.

a: Arya plays video games

b: Bran plays video games

c: Catelyn plays video games

d: Davos plays video games

e: Eddard plays video games

(1) Either Arya or Bran, or both, want to play video games: $(a \vee b) \vee (a \wedge b)$

(2) Catelyn and Davos disagree on what they want to play: $(c \wedge \neg d) \vee (\neg c \wedge d)$

(3) If Eddard plays video games, then so does Catelyn: $e \rightarrow c$

(4) Arya and Davos will play the same game: $(a \wedge d) \vee (\neg a \wedge \neg d)$

(5) If Bran plays video games, then so do Arya and Eddard: $b \rightarrow (a \wedge e)$

Assume: b is true

Looking at (5): $b \rightarrow (a \wedge e)$, therefore a is true and e is true.

Looking at (3): $e \rightarrow c$, c is true since e is true.

Looking at (4): $(a \wedge d) \vee (\neg a \wedge \neg d)$, if a is true then d has to be true.

Looking at (2): $(c \wedge \neg d) \vee (\neg c \wedge d)$, c and d can't both be true. This is a contradiction and b will be false. Therefore, Bran is playing basketball.

Looking at (1): $(a \vee b) \vee (a \wedge b)$, since b is false a has to be true

Looking at (4): $(a \wedge d) \vee (\neg a \wedge \neg d)$, since a is true, then d is true.

Looking at (2): $(c \wedge \neg d) \vee (\neg c \wedge d)$, since d is true the c is false.

Looking at (3): $e \rightarrow c$, if c is false then e has to be false to keep the implication true.

Therefore, Arya and Davos are playing video games while Bran, Catelyn, and Eddard are playing basketball.

4. **[6 Pts]** Determine whether $((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$ is a tautology using truth tables.

p	q	r	$p \rightarrow q$	$q \rightarrow r$	$p \rightarrow r$	$(p \rightarrow q) \wedge (q \rightarrow r)$	$((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$
T	T	T	T	T	T	T	T
T	T	F	T	F	F	F	T
T	F	T	F	T	T	T	T
T	F	F	F	T	F	T	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	F	F
F	F	T	T	T	T	T	T
F	F	F	T	T	T	T	T

$((p \rightarrow q) \wedge (q \rightarrow r)) \leftrightarrow (p \rightarrow r)$ is not a tautology, since it is not true for all cases.

5. [6 Pts] Prove that $(p \rightarrow (q \rightarrow r))$ and $(q \rightarrow (p \rightarrow r))$ are logically equivalent by deduction using a series of logical equivalences studied in class (truth tables will not be allowed).

Expressions 1: $(p \rightarrow (q \rightarrow r))$

Implication Rule 1: $q \rightarrow r \equiv \neg q \vee r$

$p \rightarrow (\neg q \vee r)$

Implication Rule 1: $p \rightarrow (\neg q \vee r) \equiv \neg p \vee (\neg q \vee r)$

Associative Law: $\neg p \vee (\neg q \vee r) \equiv (\neg p \vee \neg q) \vee r$

$(\neg p \vee \neg q) \vee r$

Expression 2: $(q \rightarrow (p \rightarrow r))$

Implication Rule 1: $p \rightarrow r \equiv \neg p \vee r$

$q \rightarrow (\neg p \vee r)$

Implication Rule 1: $q \rightarrow (\neg p \vee r) \equiv \neg q \rightarrow (\neg p \vee r)$

Associative Law: $\neg q \vee (\neg p \vee r) \equiv (\neg q \vee \neg p) \vee r$

Commutative Law: $(\neg q \vee \neg p) \vee r \equiv (\neg p \vee \neg q) \vee r$

$(\neg p \vee \neg q) \vee r$

Therefore, $(p \rightarrow (q \rightarrow r)) \equiv (q \rightarrow (p \rightarrow r))$ are logically equivalent.

6. [10 Pts] The proposition $p \text{ NOR } q$ is logically equivalent to $\neg(p \vee q)$. Prove that $\{\text{NOR}\}$ is functionally complete, i.e., any propositional formula is equivalent to one whose only connective is NOR. You may assume any logical equivalences we studied in class and the fact that any formula is equivalent to some formula in DNF. You may also assume that $\{\neg, \wedge, \vee\}$ is functionally complete.

Negation can be expressed by taking the NOR of a proposition with itself: $\neg p = p \text{ NOR } p$

$$p \text{ NOR } p = \neg(p \vee p) = \neg p$$

Disjunction can be expressed by NOR-ing the NOR of p and q with itself: $p \vee q = \neg(p \text{ NOR } q) = (p \text{ NOR } q) \text{ NOR } (p \text{ NOR } q)$

Conjunction can be derived using De Morgan's Law: $p \wedge q = \neg(\neg p \vee \neg q)$

Using only NOR: $\neg p = p \text{ NOR } p$, $\neg q = q \text{ NOR } q$

$$p \wedge q = ((p \text{ NOR } p) \text{ NOR } (q \text{ NOR } q)) \text{ NOR } ((p \text{ NOR } p) \text{ NOR } (q \text{ NOR } q))$$

7. [10 Pts] Consider the proposition $(p \vee q) \leftrightarrow (\neg q \vee r)$.

(a) Construct a truth table for the proposition above.

p	q	r	$\neg q$	$p \vee q$	$\neg q \vee r$	$(p \vee q) \leftrightarrow (\neg q \vee r)$
T	T	T	F	T	T	T
T	T	F	F	T	F	F
T	F	T	T	T	T	T
T	F	F	T	T	T	T
F	T	T	F	T	T	T
F	T	F	F	T	F	F
F	F	T	T	F	T	F
F	F	F	T	F	T	F

(b) A proposition is said to be in CNF (conjunctive normal form) if it is a conjunction (and) of one or more clauses, where each clause is a disjunction (or) of basic propositions or their negations. For example, $(p \vee \neg q) \wedge (\neg p \vee \neg q \vee r) \wedge (q \vee r)$ is in CNF. Any compound proposition is also logically equivalent to some proposition in CNF. Come up with a procedure to construct a CNF proposition that is equivalent to the proposition above, starting with the truth table (show your work). **Explain why your procedure is correct.**

In order to construct a CNF, we need to first look at where the formula is false. Looking at the truth table, the formula is false at row 2 ($p = T, q = T, r = F$), row 6 ($p = F, q = T, r = F$), row 7 ($p = F, q = F, r = T$), and row 8 ($p = F, q = F, r = F$). Then we create a disjunctive clause to ensure that the CNF formula is false for the combinations above and true for every other combination.

Row 2: $p = T, q = T, r = F$

$\neg p \vee \neg q \vee r$

Row 6: $p = F, q = T, r = F$

$p \vee \neg q \vee r$

Row 7: $p = F, q = F, r = T$

$p \vee q \vee \neg r$

Row 8: $p = F, q = F, r = F$

$p \vee q \vee r$

Final CNF: $(\neg p \vee \neg q \vee r) \wedge (p \vee \neg q \vee r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$

Through this procedure, we obtained a CNF formula that is logically equivalent to the original proposition.

8. [10 Pts] For the following problems, let $S(x)$, $F(x)$ and $A(x, y)$ be the statements “ x is a student”, “ x is a faculty member” and “ x has asked y a question”. Let the domain be all people at ISU. Translate the following sentences into logic.

Given:

$S(x) = x$ is a student

$F(x) = x$ is a faculty member

$A(x, y) = x$ has asked y a question

- (a) There are at least two faculty members who have not asked questions to any students.

x will be faculty member 1, y will be faculty member 2, and z will be a student.

There are at least two faculty members: $\exists x \exists y$

Those members are distinct/different: $(x \neq y)$

Assert that x and y are faculty: $F(x) \wedge F(y)$

Who have not asked questions to any students for x : $\forall (S(z) \rightarrow \neg A(x, z))$

Who have not asked questions to any students for y : $\forall (S(z) \rightarrow \neg A(y, z))$

Final Answer:

$\exists x \exists y (x \neq y \wedge F(x) \wedge F(y) \wedge (\forall (S(z) \rightarrow \neg A(x, z))) \wedge (\forall (S(z) \rightarrow \neg A(y, z))))$

- (b) Everyone has asked a question to at least one student and at least one faculty member.

x is a person at ISU, y is a student, and z is a faculty member

Everyone: $\forall x$

has asked a question to at least one student: $\exists y (S(y) \wedge A(x, y))$

and at least one faculty member: $\exists z (F(z) \wedge A(x, z))$

Final Answer:

$\forall x ((\exists y (S(y) \wedge A(x, y))) \wedge (\exists z (F(z) \wedge A(x, z))))$

9. [15 Pts] Some of the presidential candidates are standing in line on stage for a debate. There are at least two candidates on stage. Translate the assertions below into predicate formulas using $\exists$ and $\forall$ and logical operators $\neg$, $\wedge$, $\vee$, $\rightarrow$ and $\leftrightarrow$. You may use the equality predicate and the predicate $left(x, y)$, meaning “ x is somewhere to the left of y ”. You can use the predicates you define first to define subsequent predicates. Assume that the universe is the set of all presidential candidates.

For partial credit, give a short explanation of your construction.

- (a) [5 Pts] $onstage(x)$ meaning ‘ x is on stage’ *Hint: x is to the left or to the right of someone else, since there are at least two.*

Answer: $onstage(x) \equiv \exists y(y \neq x \wedge (left(x, y) \vee left(y, x)))$

Explanation: $left(x, y)$ means that x is to the left of y while $left(y, x)$ means that x is to the right of y . $y \neq x$ ensures that we are talking about someone else on the stage since there are at least two candidates. This means that if x is to the right or left of some candidate then that means that x is on the stage.

- (b) [5 Pts] $leftmost(x)$ meaning ‘ x is leftmost in line’

Answer: $leftmost(x) \equiv \forall y(y \neq x \rightarrow \neg left(y, x))$

Explanation: The predicate $leftmost(x)$ means that x is the candidate furthest to the left, so no one else is to the left of x . To express this formally, we say that for every other candidate y (where $y \neq x$), it is not true that y is to the left of x . This ensures that x is not preceded by anyone in the line, making x the leftmost.

- (c) [5 Pts] $second(x)$ meaning ‘ x is second from the left in line’

Answer: $second(x) \equiv \exists y(y \neq x \wedge left(y, x) \wedge leftmost(y) \wedge \forall z(left(z, x) \rightarrow z = y))$

Explanation: The predicate $second(x)$ means that x is second from the left, so there must be exactly one person to the left of x and that person must be the leftmost. We express this by saying that there exists a candidate y that is different from x , is to the left of x , and is the leftmost. Then we ensure that anyone else who is to the left of x must actually be y , meaning y is the only person to the left of x .

- (d) **Extra Credit** [5 Pts] $isnext(x, y)$ meaning ‘ x is immediately to the left of y in line’

10. [15 Pts] Define predicates and prove the following using the appropriate rules of inference:

- (a) [5 Pts] John, a Computer Science student, has studied logic. Everyone who studies logic is good at arguments. Therefore, there is a Computer Science student who is good at arguments.

Universe: all ISU students

Part of the solution for this sub-question has been solved for you. Fill the remaining spaces. Follow similar structure for (b) and (c), as well

Solution:

Let, $C(x) : x$ is a Computer Science student;

$L(x) : x$ studied logic;

$A(x) : x$ is good at arguments

- | | |
|--|--------------------------------|
| 1) $C(\text{John}) \wedge L(\text{John})$ | Premise |
| 2) $\forall x(L(x) \rightarrow A(x))$ | Premise |
| 3) $L(\text{John}) \rightarrow A(\text{John})$ | Universal Instantiation (UI) 2 |
| 4) $L(\text{John})$ | Simplification 1 |
| 5) $A(\text{John})$ | Modus Ponens 3,4 |
| 6) $C(\text{John})$ | Simplification 1 |
| 7) $C(\text{John}) \wedge A(\text{John})$ | Conjunction 6 and 5 |
| 8) $\exists x(C(x) \wedge A(x))$ | Existential Generalization 7 |

- (b) [5 Pts] All action movies are popular. There is an action movie filmed in Iowa. Therefore, some popular movie was filmed in Iowa.

Universe: all movies

Let, $A(x) : x$ is an action movie;

$P(x) : x$ is popular;

$I(x) : x$ is was filmed in Iowa;

$s : x$ is specific movie;

- | | |
|---------------------------------------|--------------------------------|
| 1) $\forall x(A(x) \rightarrow P(x))$ | Premise |
| 2) $\exists x(A(x) \wedge I(x))$ | Premise |
| 3) $A(s) \wedge I(s)$ | Universal Instantiation (UI) 2 |
| 4) $A(s)$ | Simplification 3 |
| 5) $I(s)$ | Simplification 3 |
| 6) $A(s) \rightarrow P(s)$ | Universal Instantiation 1 |
| 7) $P(s) \wedge A(\text{John})$ | Modus Ponens 4 and 6 |
| 8) $P(s) \wedge I(s)$ | Conjunction 7 and 5 |
| 9) $\exists x(P(x) \wedge I(x))$ | Existential Generalization 8 |

- (c) [5 Pts] Anyone who lives in Hawaii lives close to the ocean. If you don't know how to surf, you don't live close to the ocean. Anyone who knows how to surf, owns a surfboard. Therefore, anyone who lives in Hawaii owns a surfboard.

Universe: all people

Let, $H(x) : x$ live in Hawaii;

$O(x) : x$ live close to the ocean;

$S(x) : x$ lives close to the ocean;

$B(x) : x$ owns a surfboard;

1) $\forall x(H(x) \rightarrow O(x))$ Premise

2) $\forall x(\neg S(x) \rightarrow \neg O(x))$ Premise

3) $\forall x(S(x) \rightarrow B(x))$ Premise

4) $\forall x(O(x) \rightarrow S(x))$ Contrapositive 2

5) $\forall x(H(x) \rightarrow S(x))$ Hypothetical Syllogism 1,4

6) $\forall x(H(x) \rightarrow B(x)) \rightarrow P(s)$ Hypothetical Syllogism 5,3

11. [6 Pts] Are $\forall x(P(x) \rightarrow Q(x))$ and $\forall xP(x) \rightarrow \forall xQ(x)$ logically equivalent? If yes, give a proof. If no, give a counterexample.

$\forall x(P(x) \rightarrow Q(x))$ and $\forall xP(x) \rightarrow \forall xQ(x)$ is not logically equivalent.

Counterexample:

Let the domain be $D = [2,3]$

Define:

x	$P(x)$	$Q(x)$
2	T	F
3	F	T

Now the first statement will be evaluated $\forall x(P(x) \rightarrow Q(x))$:

$x=2$:

$P(2) \rightarrow Q(2) = \text{true} \rightarrow \text{false} = \text{false}$

$x=3$:

$P(3) \rightarrow Q(3) = \text{false} \rightarrow \text{true} = \text{true}$

Since not all implications are all true or all false, the first statement is false

Now the second statement will be evaluated $\forall xP(x) \rightarrow \forall xQ(x)$:

$\forall P(x)$:

$P(2) = \text{true}, P(3) = \text{false} \rightarrow \text{So } \forall P(x) = \text{false}$

$\forall Q(x)$:

$Q(2) = \text{false}, Q(3) = \text{true} \rightarrow \text{So } \forall Q(x) = \text{false}$

Since all the implications are false, the second statement is true.

Since statement 1 is false and statement 2 is true, $\forall x(P(x) \rightarrow Q(x))$ and $\forall xP(x) \rightarrow \forall xQ(x)$ is not logically equivalent.

For more practice, you are encouraged to work on the problems given at the end of Rosen, Sections 1.1 - 1.2, 1.4 - 1.6 and LLM Chapter 3.